Software Requirements Specification

SMART COMPLAINT MANAGEMENT SYSTEM

VERSION 1.0

GROUP 8

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Revision History

Name	Date	Reason For Changes	Version
Group 8	13 September 2026	Initial version of the SRS	1.0

1. Introduction

1.1 Purpose

The purpose of this project is to develop a Smart Complaint Management System (SCMS) that provides a centralized platform for users to submit, track and manage complaints. The system enables administrators to review complaints, update their status, provide resolutions and close complaints in a structured manner.

1.2 Problem Statement

Traditional complaint handling often relies on manual processes, making it difficult for users to track complaints and for administrators to manage them efficiently. The proposed SCMS provides a centralized system where complaints can be securely submitted, tracked through their lifecycle and resolved with proper status updates and responses.

1.3 Objectives

The main objectives of the proposed system are:

- To provide users with a simple platform for registering and submitting complaints.
- To generate a unique complaint ID for every successfully submitted complaint.
- To allow users to track the current status and updates of their complaints.
- To provide administrators with a centralized interface for reviewing and managing complaints.
- To allow administrators to update complaint status and provide resolutions.
- To maintain complaint information securely throughout its lifecycle.

1.4 Scope

The Smart Complaint Management System focuses on the complete complaint lifecycle, from user registration and complaint submission to tracking, administrative processing, resolution and closure.

The system covers:

- User registration, authentication and logout.
- Complaint submission and validation.
- Unique complaint ID generation.
- Complaint storage and retrieval.
- Complaint status tracking and updates.
- Administrator login and complaint management.
- Complaint resolution and closure.
- Display of final resolution and status to the concerned user.

1.5 References

[1] P. Pathmanathan and R. Poulier, "A Comprehensive Citizen Engagement Framework for Effective Resolution of Public Complaints in Cities," IEEE International Conference on Smart Technologies for Smart Nation (SmartTechCon), 2017, DOI: 10.1109/SmartTechCon.2017.8358545.

[2] "Smart Student Grievance Redressal System with Foul Language Detection," 2022 8th International Conference on Advanced Computing and Communication Systems (ICACCS), 2022, DOI: 10.1109/ICACCS54159.2022.9785119.

1.6 Audience

This document is intended for the project development team, testers, administrators, users, faculty members and project evaluators. It provides a common understanding of the functional requirements, non-functional requirements, system behaviour and expected functionality of the Smart Complaint Management System.

2. Overall Description

2.1 Product Perspective

The Smart Complaint Management System (SCMS) is a centralized platform for submitting, tracking, managing, and resolving complaints. The system connects users and administrators through a structured complaint lifecycle, from registration and complaint submission to status updates, resolution, and closure.

2.2 Major Product Functions

The major functions of the system include user registration and authentication, complaint submission, generation of a unique complaint ID, complaint storage, complaint tracking, administrator login, complaint review, status updates, resolution or response, and complaint closure. The final resolution and status are made available to the concerned user.

2.3 User Classes and Characteristics

The system primarily consists of two user classes: Users and Administrators. Users can register, submit complaints, view their complaints, track their status, and access updates and final resolutions. Administrators are authorized personnel who can access submitted complaints, review complaint details, update their status, provide responses or resolutions, and close complaints.

2.4 Operating Environment

The Smart Complaint Management System is intended to operate as a software-based application accessible through a user interface. The system requires a suitable computing environment, database support for storing user and complaint information, and network connectivity for communication between users, administrators, and the application services.

2.5 Constraints

The system must ensure that protected functions are accessible only to authenticated users and authorized administrators. User passwords must not be stored in plain text, and users must not be allowed to view or modify

complaints belonging to other users. Important administrator actions related to complaint status and resolution must also be recorded.

2.6 Assumptions and Dependencies

The system assumes that users provide the required information while registering and submitting complaints. Administrators are assumed to have authorized access to administrative functions and are responsible for reviewing and processing complaints. The system depends on reliable storage and availability of complaint information so that successfully submitted complaints can be tracked and processed throughout their lifecycle.

2.7 System-Wide Requirements

The system shall maintain common requirements across all modules, including performance, availability, usability, security, reliability, and maintainability. Normal user requests should receive a response within 3 seconds, the system should remain available during normal operating periods, sensitive information must be protected, and the system should be organized into independent modules for easier maintenance.

3. External Interface Requirements

3.1 User Interface

The system shall provide a simple and user-friendly interface for users and administrators to register, submit, track, manage, update, and resolve complaints.

3.2 Software Interface

The system shall interact with a database for storing user details, complaint information, status updates, and resolution records.

3.3 Communication Interface

The system shall use network communication to allow users and administrators to access the complaint management system and exchange required information.

3.4 Administrator Interface

The system shall provide authorized administrators with an interface to view complaints, update their status, provide resolutions, and close complaints.

3.5 User Interface

Users shall be able to access their submitted complaints, track their current status, view updates, and see the final resolution.

4. System Features

4.1 User Account & Complaint Submission

Description and Priority

Users can register an account, log in to the system and submit complaints by providing the required details. The system validates the complaint information, generates a unique complaint ID and stores the complaint. Priority: High.

Stimulus/Response

A user fills in the registration details and submits them; the system creates the account. A logged-in user enters the required complaint details and submits the complaint; the system validates the information, generates a unique complaint ID and stores the complaint.

Functional Requirements

- SCMS-F-001: The system shall allow a new user to register using the required details.
- SCMS-F-002: The system shall authenticate registered users using valid login credentials.
- SCMS-F-003: The system shall allow an authenticated user to log out.
- SCMS-F-004: The system shall allow an authenticated user to lodge a complaint with the required details.
- SCMS-F-005: The system shall generate a unique complaint ID for every successfully lodged complaint.
- SCMS-F-006: The system shall store the details of every successfully lodged complaint.
- SCMS-F-007: The system shall validate mandatory complaint details before submission.

4.2 Complaint Tracking and User Status

Description and Priority

Users can view the complaints they have submitted, track their current status and view updates or responses related to their complaints. The system provides the final resolution and status to the concerned user. Priority: High.

Stimulus/Response

A user opens their complaint list and selects a complaint; the system displays its current status and available updates. When the complaint is resolved, the system displays the final resolution and status to the user.

Functional Requirements

- SCMS-F-008: The system shall allow users to view complaints they have submitted.
- SCMS-F-009: The system shall allow users to track the current status of a submitted complaint.
- SCMS-F-010: The system shall display the current status of each complaint to the concerned user.
- SCMS-F-011: The system shall display updates or responses associated with the user's complaint.
- SCMS-F-018: The system shall make the complaint resolution and final status available to the concerned user.

4.3 Admin Complaint Management

Description and Priority

Administrators can securely log in to the system, view submitted complaints, access complaint details and update the status of complaints while they are being processed. Priority: High.

Stimulus/Response

An administrator logs in to the admin portal; the system authenticates the administrator. The administrator selects a complaint, views its details and updates its status according to the progress of the complaint.

Functional Requirements

- SCMS-F-012: The system shall allow an authorized administrator to log in to administrative functions.
- SCMS-F-013: The system shall allow the administrator to view submitted complaints.
- SCMS-F-014: The system shall allow the administrator to view details of a selected complaint.
- SCMS-F-015: The system shall allow the administrator to update the status of a complaint.

4.4 Complaint Resolution and Closure

Description and Priority

Administrators can provide a resolution or response for a complaint and close the complaint after the issue has been addressed. The final outcome is made available to the concerned user. Priority: Medium.

Stimulus/Response

An administrator enters a resolution or response for a complaint and updates its status after processing. The system records the resolution, closes the complaint and makes the final status available to the concerned user.

Functional Requirements

- SCMS-F-016: The system shall allow the administrator to provide a resolution or response for a complaint.
- SCMS-F-017: The system shall allow the administrator to close a complaint after resolution.
- SCMS-F-018: The system shall make the complaint resolution and final status available to the concerned user.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Since the Smart Complaint Management System runs on our local setup for this version, we are keeping the performance targets realistic:

- Common actions such as logging in, submitting a complaint, viewing complaints and checking complaint status should complete within about 2–3 seconds under normal use.
- When a user submits a complaint, the complaint should be validated, stored and a complaint ID should be generated within about 2 seconds.
- The complaint status and updates page should load within about 3 seconds.

- The system should comfortably handle around 20–30 users at a time, which matches a typical academic or small organizational deployment.

5.2 Safety Requirements

This system does not control any physical process and therefore does not introduce specific physical safety hazards. As a basic safeguard, complaint information should be stored in the database rather than only in temporary memory to reduce the risk of permanent data loss.

5.3 Security Requirements

- Users must register and log in before they can submit or access their complaints.
- Passwords shall be stored in hashed form and not as plain text.
- Users shall only be allowed to access their own complaint information.
- Admin-only functions such as viewing and managing submitted complaints and updating complaint status shall be restricted to authorized administrators.
- The system shall validate user input to prevent invalid or unauthorized data from being stored.

5.4 Software Quality Attributes

- Usability: The interface should be simple and clear so that users can easily register, submit complaints, track their status and view resolutions with minimal navigation.
- Reliability: Complaint details, status updates and resolutions should be stored correctly and should not be lost during normal system usage or page refreshes.
- Maintainability: The system should use a modular structure with separate front-end, back-end and database components so that individual parts can be modified or extended easily.
- Scalability: The design should allow additional users and complaints to be handled in the future without requiring major changes to the core system.

5.5 Business Rules

- Only registered and logged-in users can submit complaints and access their complaint information.
- A user can view and track only the complaints associated with their own account.
- Only authorized administrators can view and manage submitted complaints and update their status.
- Only administrators can provide the final resolution and close a complaint.
- Complaint resolution and final status shall be made available to the concerned user after the complaint is resolved.

5.6 Detailed Security Requirements

5.6.1 Security Objectives

The main security objectives of the Smart Complaint Management System are:

- To protect user accounts and complaint information from unauthorized access.
- To ensure that only authenticated users can access complaint-related services.

- To restrict administrative functions to authorized administrators only.
- To maintain the confidentiality and integrity of user and complaint data.

5.6.2 Security Requirements

- SCMS-NFR-SEC-001: The system shall require users to register and log in before accessing complaint-related services.
- SCMS-NFR-SEC-002: The system shall store user passwords in hashed form and shall not store passwords as plain text.
- SCMS-NFR-SEC-003: The system shall allow users to view and manage only their own complaint information.
- SCMS-NFR-SEC-004: The system shall restrict administrative functions such as viewing all complaints, updating complaint status, providing resolutions, and closing complaints to authorized administrators.
- SCMS-NFR-SEC-005: The system shall validate user inputs before storing complaint and account information to reduce invalid or malicious data entry.
- SCMS-NFR-SEC-006: The system shall prevent unauthorized users from accessing protected pages and functions.
- SCMS-NFR-SEC-007: The system shall maintain the integrity of complaint records so that complaint details and status updates cannot be modified by unauthorized users.

6. Quality Attributes & Acceptance Tests

6.1 Quality Attributes

The Smart Complaint Management System shall satisfy the following quality attributes:

- Usability: The system should provide a simple and clear interface that allows users to submit complaints, track their status and view resolutions easily.
- Reliability: Complaint details, status updates and resolutions should be stored correctly without being lost during normal system usage.
- Maintainability: The system should follow a modular structure so that individual components can be modified or extended easily.
- Scalability: The system should support an increasing number of users and complaints without requiring major changes to the core system.
- Security: User information and complaint data should be protected through authentication, authorization and secure data handling.

6.2 Acceptance Tests

Test Case ID	Requirement	Acceptance Criteria
TC-REG-01	SCMS-F-001	A new user can successfully register using valid details.
TC-AUTH-01	SCMS-F-002	A registered user can log in using valid credentials.

Test Case ID	Requirement	Acceptance Criteria
TC-CMP-01	SCMS-F-004	An authenticated user can successfully lodge a complaint.
TC-CMP-02	SCMS-F-005	A unique complaint ID is generated for every successfully lodged complaint.
TC-TRK-01	SCMS-F-008	A user can view complaints submitted through their account.
TC-TRK-02	SCMS-F-009	A user can track the current status of a submitted complaint.
TC-ADM-01	SCMS-F-012	An authorized administrator can log in to administrative functions.
TC-ADM-02	SCMS-F-013	An administrator can view submitted complaints.
TC-ADM-04	SCMS-F-015	An administrator can update the status of a complaint.
TC-RES-01	SCMS-F-016	An administrator can provide a resolution or response.
TC-RES-02	SCMS-F-017	An administrator can close a complaint after resolution.
TC-RES-03	SCMS-F-018	The concerned user can view the final resolution and status.

Acceptance Criteria

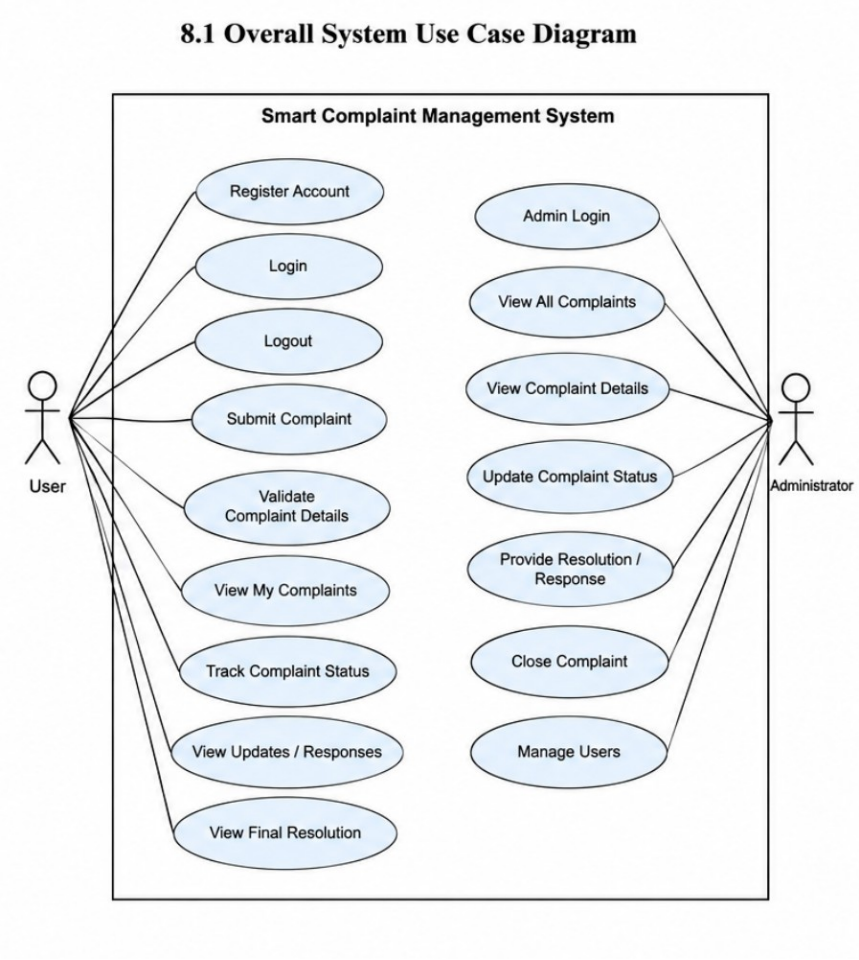
The system will be considered acceptable when all high-priority functional requirements are implemented and successfully verified, complaint data is stored correctly, authorized users can access the appropriate functions, and no critical security or functional issues remain.

8. System Models and Diagrams

8.1 Overall System Use Case Diagram

The overall use case diagram represents the interaction between the two primary actors, User and Administrator, and the major functions of the Smart Complaint Management System.

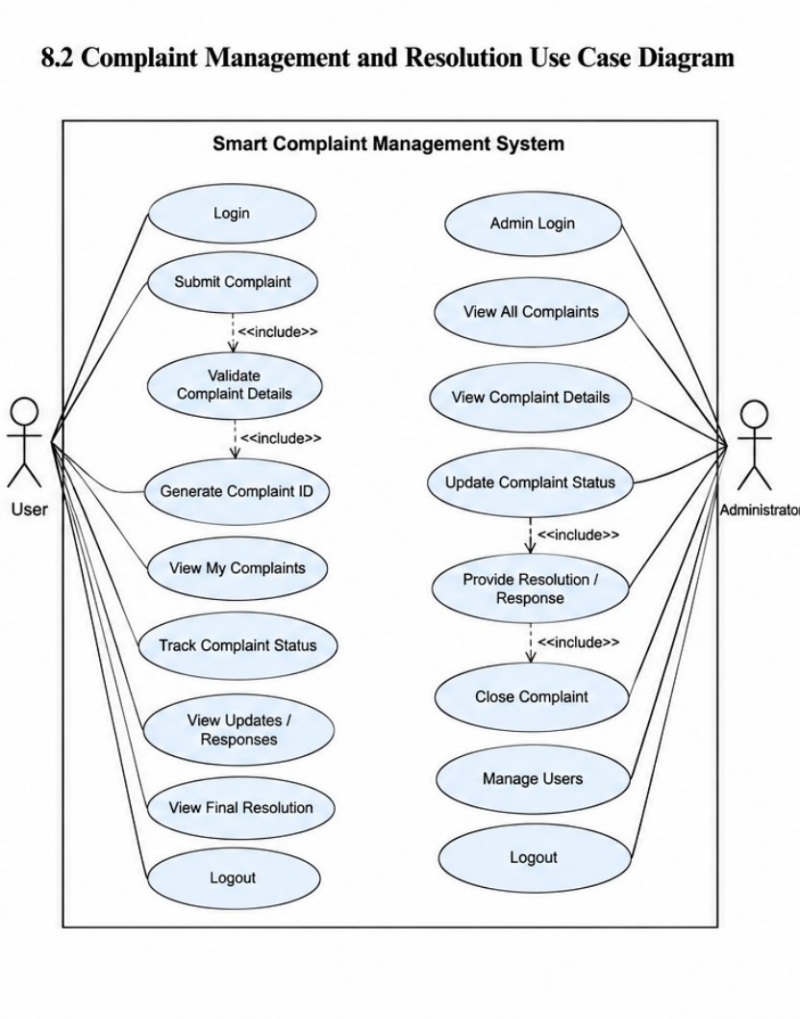
8.1 Overall System Use Case Diagram



8.2 Complaint Management and Resolution Use Case Diagram

The complaint management use case diagram represents the detailed interactions involved in complaint submission, tracking, administrative processing, status updates, resolution and closure.

8.2 Complaint Management and Resolution Use Case Diagram



9. Requirements Traceability Matrix

9.1 Requirements Traceability Matrix

Req ID	Requirement	Module	Test Case	Status
SCMS-F-001	User registration	User Account & Complaint Submission	TC-REG-01	N
SCMS-F-002	User authentication	User Account & Complaint Submission	TC-AUTH-01	N
SCMS-F-003	User logout	User Account & Complaint Submission	TC-AUTH-02	N
SCMS-F-004	Lodge complaint	User Account & Complaint Submission	TC-CMP-01	N
SCMS-F-005	Generate unique complaint ID	User Account & Complaint Submission	TC-CMP-02	N
SCMS-F-006	Store complaint details	User Account & Complaint Submission	TC-CMP-03	N
SCMS-F-007	Validate complaint details	User Account & Complaint Submission	TC-CMP-04	N
SCMS-F-008	View submitted complaints	Complaint Tracking	TC-TRK-01	N
SCMS-F-009	Track complaint status	Complaint Tracking	TC-TRK-02	N
SCMS-F-010	Display complaint status	Complaint Tracking	TC-TRK-03	N
SCMS-F-011	Display complaint updates	Complaint Tracking	TC-TRK-04	N
SCMS-F-012	Admin login	Admin Complaint Management	TC-ADM-01	N
SCMS-F-013	Admin views complaints	Admin Complaint Management	TC-ADM-02	N
SCMS-F-014	Admin views complaint details	Admin Complaint Management	TC-ADM-03	N
SCMS-F-015	Admin updates complaint status	Admin Complaint Management	TC-ADM-04	N
SCMS-F-016	Admin provides resolution	Complaint Resolution &	TC-RES-01	N

Req ID	Requirement	Module	Test Case	Status
		Closure		
SCMS-F-017	Admin closes complaint	Complaint Resolution & Closure	TC-RES-02	N
SCMS-F-018	User views final resolution	Complaint Resolution & Closure	TC-RES-03	N

10. Other Requirements

All application data, including user accounts, complaints, complaint IDs, complaint status, resolutions and related records, will be stored in the database. GitHub will be used for version control and Jira will be used to track development tasks and sprints. No payment or internationalisation features are planned for this version.

Appendix A: Glossary

SCMS: Smart Complaint Management System – the platform used to submit, track, manage and resolve complaints.

Complaint: A request or issue submitted by a user through the system for review and resolution.

Complaint ID: A unique identifier generated for each complaint, used by the user to track its status.

Complaint Status: The current stage of a complaint, such as Submitted, Under Review, In Progress, Resolved or Closed.

Resolution: The action or response provided by the administrator to address and resolve a complaint.

User: A registered person who can submit complaints, track their status and view the resolution.

Administrator (Admin): An authorized person who can view, manage, update and resolve complaints submitted by users.